

Module 03: Perception — Predictive Coding in the Brain

Course: Active Inference 101 | **Unit:** Computational Neuroscience | **Audience:** First-semester undergraduates

Learning Objectives

1. Describe the **predictive coding** model: how cortical layers generate predictions and compute prediction errors.
2. Explain the **canonical microcircuit** — the basic neural wiring pattern that implements predictive coding.
3. Identify neural signatures of prediction and prediction error in real brain data (e.g., mismatch negativity).

Introduction

The Cognitive Science unit introduced perception as prediction. This module asks: *how does the brain actually do it?* The answer is **predictive coding** — a specific computational architecture implemented in the layers and connections of the cerebral cortex.

Key Concepts

1. The Predictive Coding Architecture

In predictive coding, the cortex implements two streams of information:

- **Top-down predictions:** Higher areas send predictions to lower areas about what they should be seeing/hearing/feeling
- **Bottom-up prediction errors:** Lower areas compute the mismatch between predictions and actual sensory data, and send these errors upward

Only the *error* travels upward. If the prediction is correct, there's nothing to report. This is extremely efficient — the brain only communicates what's unexpected.

2. Cortical Layers and the Canonical Microcircuit

The cortex has six layers, and different layers have specific roles in predictive coding:

- **Superficial layers (layers 2/3):** Encode prediction errors and send them upward to higher areas
- **Deep layers (layers 5/6):** Encode predictions and send them downward to lower areas
- **Layer 4:** Receives input (from the thalamus or from higher areas)

This **canonical microcircuit** is repeated throughout the cortex. The same basic wiring pattern — predictions flowing down, errors flowing up — is found in visual cortex, auditory cortex, somatosensory cortex, and beyond.

3. Neural Evidence: The Mismatch Negativity (MMN)

One of the strongest pieces of evidence for predictive coding comes from the **mismatch negativity (MMN)**:

- Play a repeated sound: "beep, beep, beep, beep..."
- Occasionally insert a different sound: "beep, beep, beep, BOOP, beep..."
- The brain produces a distinctive electrical response (the MMN) to the unexpected sound

The MMN is a **prediction error signal** — the brain predicted "beep" and got "boop." This response is automatic (occurs even during sleep) and is generated in the auditory cortex. It's one of the clearest demonstrations that the brain is constantly predicting.

4. Precision and Gain Control

In neural terms, **precision** is implemented through **gain control** — adjusting the sensitivity (gain) of neural populations:

- **High precision** = high gain = prediction errors are amplified (the brain is "listening closely")

- **Low precision** = low gain = prediction errors are suppressed (the brain is "ignoring" this channel)

Neuromodulators like acetylcholine adjust gain in sensory cortex, while dopamine adjusts gain in subcortical circuits. Attention is literally the process of turning up the gain on relevant neural populations.

5. Visual Processing as Predictive Coding

Vision provides the best-studied example of predictive coding:

- V1 (primary visual cortex) processes edges and orientations
- V2 processes textures and patterns
- V4 processes color and shape
- IT (inferotemporal cortex) processes objects and faces

In predictive coding, higher areas (V4, IT) predict what lower areas (V1, V2) will see. V1 only sends upward the prediction errors — the unexpected parts of the visual scene. This is why you rarely notice the stable parts of your visual field but immediately notice anything that changes.

Summary

Predictive coding is implemented in the cortex through a canonical microcircuit: deep layers generate top-down predictions while superficial layers compute bottom-up prediction errors. Neural evidence like the mismatch negativity confirms that the brain constantly predicts sensory input. Precision is implemented through gain control — adjusting how loudly prediction errors "speak."

Further Reading

- Rao, R. P. N. & Ballard, D. H. (1999). Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1), 79-87.
- Bastos, A. M. et al. (2012). Canonical microcircuits for predictive coding. *Neuron*, 76(4), 695-711.

- Garrido, M. I. et al. (2009). The mismatch negativity: A review of underlying mechanisms. *Clinical Neurophysiology*, 120(3), 453-463.